APPENDIX I: MADAGASCAR—JOINT BANK-FUND DEBT SUSTAINABILITY ANALYSIS

Madagascar's risk of debt distress is low. The upgrading of Madagascar's risk of debt distress rating compared to the previous debt sustainability analysis (DSA) is on account of the improved growth and export prospects linked to the big mining projects under construction. Madagascar's external debt burden indicators under the baseline scenario remain below the relevant policy-dependent indicative thresholds. The stress tests do not result in indicators breaching the thresholds although they reveal that external debt sustainability is sensitive to export shocks, which are likely given the concentration of Madagascar exports. The inclusion in the analysis of the small domestic debt does not change the assessment of country's risk of debt distress.

A. Introduction

1. **This DSA has been prepared jointly by IMF and World Bank staff.** It is based on the framework for low-income countries approved by the respective Executive Boards. The framework takes into account indicative thresholds for debt burden indicators determined by the quality of the country's policies and institutions.[1] It comprises of a baseline scenario (which assumes, among others, full delivery of Heavily Indebted Poor Countries (HIPC) Initiative debt relief by external creditors)[2] and a set of alternative scenarios.

B. Recent Developments and Current Debt Situation

2. **Madagascar reached its completion point under the Enhanced HIPC Initiative in October 2004.** The resulting debt relief, including additional bilateral debt relief from most Paris Club creditors, reduced the end-2003 NPV of debt-to-exports ratio to an estimated 137 percent.[3] [4]At that time, the ratio was projected to increase to 154 percent in 2004 and decline thereafter.

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[1] According to the World Bank Country and Policy Institutional Assessment (CPIA) Index, Madagascar is rated as a medium performer. This rating is the same using either the latest index or the three-year average. The indicative sustainability thresholds for external debt applicable for that category of countries are: (i) 150 percent for NPV of debt-to-exports ratio; (ii) 40 percent for the NPV of debt-to-GDP ratio; (iii) 250 percent for the NPV of debt to fiscal revenues ratio; (iv) 20 percent for the debt service to exports ratio; and (v) 30 percent for the debt service to revenue ratio.

[2] As of end-2006, Madagascar has estimated US$595 million in arrears towards non-Paris Club and private creditors that are not delivering HIPC debt relief (Algeria, Libya, and Iraq are the largest accounting for 92 percent of the total). Madagascar continues to make efforts to contact these creditors, regularize the payments, and obtain full debt relief under the HIPC Initiative.

[3] Total debt relief to Madagascar under the Initiative amounts to US$836 million in NPV terms.

[4] Most Paris Club creditors have written-off their outstanding claims after full delivery of HIPC debt relief, reducing claims on Madagascar by further US$466 million in NPV terms. In 2006, Sweden and Israel canceled 100 percent of the Malagasy debt. In addition an agreement was finalized with Japan and an agreement is under discussion with the Federation of Russia. Regarding non-Paris Club creditors, cancellation agreements should be finalized soon with Abu Dhabi and the People's Republic of China.

Madagascar : External Debt Outstanding 2006

Creditor	Amounts (millions of $US)	In percent of GDP	Share (in percent)
Total external debt	1,584.4	28.8	100
Bilateral Creditors	724.5	13.1	46
Paris Club	111.6	2.0	7
Other countries	612.9	11.1	39
Private creditors	14.3	0.3	1
Multilateral	845.6	15.3	53

Source: Madagascar authorities and Fund Staff estimates

3. **Madagascar's external public debt declined significantly from US$3.5 billion at end 2005 (including arrears) to US$1.6 billion at end 2006 on the account of the debt relief under the Multilateral Debt Relief Initiative (MDRI).** The IMF's debt relief under the MDRI became effective in January 2006 and amounts to US$197 million.[5] MDRI debt relief from IDA was implemented starting in July 2006 through the 100 percent reduction of the debt disbursed as of end-2003 that was still outstanding at the implementation date. Total MDRI debt service relief from IDA is estimated at SDR1.2 billion (equivalent to about US$1,780.1 million).[6] As for the ADF, the delivery of debt relief under the MDRI though a write-off of the outstanding obligations as of end-December 2004 became effective in September 2006 (with retroactive application to January 2006). Estimated MDRI debt relief from the ADF amounts to US$393 million. This has been partly offset by new borrowings as well as exchange rate developments[7], and the actual outstanding stock of debt has declined by 55 percent in 2006.

[5] See www.imf.org for details on the implementation of the MDRI by the IMF. The amount of relief includes undisbursed HIPC assistance from the Fund, previously expected to be delivered over time, and MDRI assistance. Excluding the HIPC share, the MDRI debt relief amounts to US$186 million.

[6] For details on the implementation of the MDRI by IDA, see IDA/SecM2005 and IDA/SecM2006-0131.

[7] This refers to the increase on the US$ value of the debt due to the depreciation of the US$ vis-à-vis other currencies in which Madagascar's debt is denominated.

4. **The MDRI debt relief has triggered a drastic change in the creditor composition.** The share of multilateral creditors has decreased from more than 75 percent at end-2005 to 53 percent at end-2006. Conversely, bilateral and commercial creditors represented 47 percent of total outstanding obligations (with Paris Club creditors accounting for 7 percent), compared to about 23 percent by end-2005. The bilateral Paris Club debt that remains is mainly to Russia. The current creditor composition is broadly similar to the one prevailing at end-2003, i.e., prior to the debt write-offs granted by most Paris Club and the MDRI debt relief.

C. Baseline Medium- and Long-Term Scenario

5. **The baseline scenario is substantially different to the one described in the previous DSA due to the impact of two large mining projects.** Construction of a US$650 million (12 percent of GDP) ilmenite mine began in mid-2006 and a US$2.5 billion (45 percent of GDP) nickel and cobalt mine and processing facility commenced in 2007 and will be completed over the next three years. Such a large inflow of capital is leading to an appreciation of the currency and the project will result in higher growth and starting in 2009/10 in higher exports (Box 1) than in the previous DSA.

6. **The baseline scenario is built on a number of key macroeconomic assumptions, the implementation of sound macroeconomic and structural policies and external financing that is through grants and highly concessional loans.** Key macroeconomic assumptions are indicated in Box 1. With respect to external financing, new borrowing is projected to remain largely at highly concessional terms. External assistance as a share of GDP, which had climbed very rapidly to close to 14 percent in 2004 due to large inflows of external aid and borrowing to finance recovery after the 2002 political crisis, will remain relatively high in the short term (increasing from 8 percent of GDP in 2005 to 11 percent of GDP in 2006, in part due to external financing of the presidential election, and 9 percent in 2007 based on recent information from donors) and gradually decrease to about 6 percent of GDP by the end of the projection period. The borrowing is largely for financing infrastructure projects aimed at lowering costs and enhancing competitiveness, and to help Madagascar attain the objectives of the Madagascar Action Plan (MAP) and the Millennium Development Goals (MDGs).

7. **Under the baseline scenario, Madagascar's external debt indicators remain well below the thresholds throughout the projection period** (Figure 1a and Table 1a). The debt indicators dropped sharply in 2006 as a result of the MDRI (Table 1a). The NPV of debt-to-GDP ratio which was at about 36 percent in 2005[8] (a level already below the threshold for medium performers) dropped to below 12 percent in 2006 (Figure 1a and Table 1a); it increases subsequently to peak at 16 percent in 2017 and drops after to about 13 percent in

[8] See IMF Country Report No. 06/306, Republic of Madagascar: Request for a Three-Year Arrangement Under the Poverty Reduction and Growth Facility and Activation of the Trade Integration Mechanism.

the outer years. The NPV of debt-to-revenue ratios (Tables 2a) exhibit a similar profile. The increase in these ratios in the initial years (from the low post-MDRI levels) reflects relatively high new borrowing albeit at concessional terms to finance needed investment expenditures. The NPV of debt-to-exports is driven by the impact of the big mining projects. Before they start their exports, these projects are expected to lead to an appreciation of the exchange rate that will negatively impact competitiveness of exports. This in conjunction with the weak performance of agricultural prices of key exports, such as vanilla, leads to a deterioration of debt indicators based on exports up to 2009. The mining projects are expected to start exporting in end-2009 or in 2010. The value of these exports is so large that the NPV of debt-to-export ratio is projected to drop from 61 percent in 2009 to 38 percent in 2010 and grow through the projected period (but remaining well below the threshold) due to new borrowing. The debt service-to-exports ratio exhibits a similar pattern. Moreover, given the highly concessional nature of the existing debt and new borrowing, the debt service ratios are well below the indicative thresholds throughout the projection period but show a rising trend due to the accumulation of new debt.

8. **Madagascar's total public debt ratios, including domestic debt, also stay well below the external debt sustainability thresholds under the base case scenario** (Figure 2a and Table 2a). The domestic debt is projected to remain under 10 percent of GDP through the projection period, as total expenditure is assumed to be broadly kept in check with available financing from tax revenue and external assistance, and the fiscal deficit is projected to remain predominantly financed by external assistance on concessional terms.

D. Sensitivity Analysis

External public debt indicators

9. **The sensitivity analysis suggests that the external debt indicators are low but could deteriorate with inappropriate policies, and/or if confronted by adverse shocks. They would however remain below the threshold.**[9]

- The debt indicators in the **historical scenario** (Scenario A1 in Figure 1A and Table 1b based on the average of the years 1999–2006) follow a lower trajectory than under the baseline. It should be noted, however, that the years leading up to 2006 include years of political turmoil, large terms of trade shocks, cyclones, and natural disasters. During that period, although real growth averaged only about 3.5 percent, the current account deficit was also low as the Export Processing Zone sector (put in place in 1989) was growing at very

[9] The stress tests performed under the sensitivity scenario assumed permanent modifications of key baseline assumptions ("alternative scenarios") as well as temporary deviations ("bound tests"). The "alternative scenarios" include a "historical scenario", under which the main variables that determine the debt dynamics are assumed to remain at their historical average, and a "financing scenario" that depicts the impact of lower concessionality in new borrowing. The "bound tests" are designed to examine the impact on debt and debt service indicators of shocks, based on the country's historical volatility, to key variables.

high rates. This factor explains the relatively benign trajectory of the debt indicators under the historic scenario. If the years of the political crisis (GDP decline of 12.7 percent in 2002) and the recovery from this (growth of 9.8 percent in 2003) are excluded, the average growth is significantly higher at 5 percent, which remains below the baseline scenario but close to it if the impact of the big mining projects is excluded.

- The **financing scenario** (Scenario A2) reveals that Madagascar's external debt indicators are sensitive to the terms of new borrowing. Under this scenario, the NPV of debt-to-GDP remains below the threshold but would more than double in the next decade (from 11 percent in 2007 to 23 percent in 2017), and remains at about that level through 2021 before decreasing slightly. The NPV of debt-to-exports ratio would increase steadily from 46 percent in 2007 to 101 percent in 2027. This increase is substantial although the ratio would remain below the threshold. The projected path of these two ratios, together with the rising trend of the debt service ratio from 2 to 6 percent during the projection period, points to risks from new borrowing at less favorable terms than assumed in the baseline scenario.[10]

- **The bound tests do not reveal major underlying vulnerabilities.** The most extreme stress tests shows the impact of the combined FDI and growth shock on the NPV of debt-to-GDP ratio (Scenario B5), and exports shock (Scenario B2) on the NPV of debt-to exports ratio do not result in indicators significantly breaching thresholds. These scenarios show that debt sustainability is most vulnerable to changes in the inflows of FDI and to the impact of these projects on growth and exports. The NPV of debt-to-GDP increases significantly (to 26 percent in 2027) but remains comfortably below the threshold level. On the other hand, the export based indicator-the NPV of debt to exports, crosses the threshold in 2009 only but falls back below the threshold as soon as the exports of the two big mining projects materialize. The shock to exports is somewhat extreme (the standard deviation is about 33 percent) as it is influenced by the rapid increase in exports in the historical period from textiles exports, and the large decline in exports that took place during the political crises and the subsequent rapid recovery. The implications of export concentration for vulnerability are nevertheless to be noted, as they illustrate the impact of shocks to certain export commodities (such as textiles), which have become large as a share of total exports in Madagascar (about 40 percent in 2005 and 2006), as well as the exports of only two mining projects that are expected to account for over 50 percent of exports of goods and services over 2011–17. However, the vulnerabilities associated with the big mining projects appear limited: even if the exports of the big mining projects are only 1/3 of their baseline level (due either to a lower volume and/or to lower export prices) the NPV of debt-to-exports would remain below the threshold (scenario B7) peaking at 71 percent in 2022 (61 percent in the baseline).

[10] This scenario is similar to a sharp decline in grant financing, compared to the baseline, compensated by loans.

Total public debt indicators

10. **Total public debt indicators are most sensitive to economic growth shocks.** A temporary or permanent deviation from the baseline real GDP growth path would bring the total public debt beyond the external debt sustainability thresholds beginning in 2015, if spending plans were kept unchanged, which is unlikely in view of past experience (Table 2b, Scenarios A3 and B1).

E. Debt Distress Classification and Conclusions

11. **Madagascar's risk of debt distress is low following the debt relief under the HIPC Initiative and the MDRI and due to the impact on exports and GDP of big mining projects.** In the baseline scenario Madagascar's debt indicators are below the thresholds. The debt situation does not appear to be significantly vulnerable to shocks. Debt sustainability appears somewhat vulnerable if large export shocks materialize. This is a risk in the medium term given the high concentration in exports in textiles in the short run and two mining products whose prices are volatile.

12. **In conclusion, the debt situation seems under control and not subject to major risks.** Sensitivity analysis nonetheless points to the need not only for implementing the policies underpinning the baseline scenario, but also for careful monitoring of borrowing policies and, more importantly, export performance. The latter is critical given the concentration of exports in textiles and clothing in the short run and in two mining projects starting in 2010. Borrowing at non-concessional terms as well new borrowing beyond the levels assumed under the baseline could also raise Madagascar's risk of debt distress.

Box 1. Baseline Macroeconomic Assumptions

Real GDP growth is projected at 7 percent over 2007-2027. This growth rate is higher than the historical average (5 percent over 1999–2006 if the impact of the political turmoil of 2002 is excluded) due to the impact of two large mining projects that started at the end of 2006. Without these projects, the real growth rates would have been a little under 6 percent on average i.e. higher but close to the historical rate.

Inflation as measured by the GDP deflator in dollar terms is projected to average about 3.7 percent, with higher rates of 20 percent in 2007 and decelerating to 2 percent subsequently. However, reflecting the impact of the large inflows of FDI on the **exchange rate**, which in turn will depend crucially of the monetary policy undertaken, the increase in the GDP deflator in local currency is expected to decelerate from 9.5 percent in 2007 to 5 percent beginning in 2010.

Export volumes grow at 11 percent on average and support GDP growth. The growth rates are relatively modest in the initial years reflecting the impact of the exchange rate appreciation on some vulnerable export industries such as textiles and shrimp but will jump as soon as the exports of the big mining projects start.

Import volumes average about 7 percent for the period. Import volumes are expected to grow significantly due to the high import content (estimated at about 80 percent) of the mining projects during their construction phase and will then slow down to about 6.5 percent i.e., at about the pace of real GDP growth.

The current account deficit is projected to increase significantly during the initial years of the projections due to the large increase in imports related to the big mining projects. However as exports of these projects start, the current account deficit would gradually shrink from about 20 percent of GDP in 2008 to 9 percent by 2010 and 5 percent by the end of the projection period.

Tax revenues are projected to increase from 10.7 percent of GDP in 2006 to about 15.5 percent of GDP in 2027, owing to the termination of ad hoc tax and import duty exemptions, tax policy reforms aimed at simplifying the corporate and personal income tax regimes, and steadfast improvements in tax and custom administration.

Total expenditures are broadly kept in check with available financing from tax revenue and external assistance, resulting in a roughly constant share of GDP over the projection period.

External assistance progressively unwinds from the exceptionally high level of about 11 percent of GDP reached during the period 2004 to 2006 (financing of recovery after the 2002 political crisis and of the 2006 presidential elections) to over 7 percent of GDP over 2007–11 (which is slightly higher than the 6 percent of GDP observed during the late 1990s because of donor financing of the MAP) and to 5 percent of GDP in outer years. The share of grants gradually increases from half in the initial years to more than half in the later years.

Table 1a. Madagascar: External Debt Sustainability Framework, Baseline Scenario, 2007-2027 [1]
(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average [6]	Standard Deviation [6]	Projections									
	2004	2005	2006			2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
External debt (nominal) [1]	79.9	69.7	28.8			26.8	26.2	27.5	28.0	28.2	29.0		29.6	23.2	
o/w public and publicly guaranteed (PPG)	79.9	69.7	28.8			26.8	26.2	27.5	28.0	28.2	29.0		29.6	23.2	
Change in external debt	-3.3	-10.2	-41.0			-2.0	-0.6	1.3	0.4	0.3	0.7		-0.2	-0.7	
Identified net debt-creating flows	25.0	-6.4	-4.8			0.6	1.1	1.5	0.6	0.8	1.3		-0.4	-2.4	
Non-interest current account deficit	7.9	10.1	8.5	5.5	2.9	17.3	20.7	19.0	8.4	8.5	8.5		6.7	4.3	6.0
Deficit in balance of goods and services	14.9	14.1	11.2			19.3	22.1	20.0	-2.6	-4.0	-3.0		-0.4	1.6	
Exports	32.6	26.9	29.7			24.7	21.8	21.2	36.1	37.0	35.4		29.2	20.8	
Imports	47.5	41.0	40.9			44.0	43.9	41.2	33.5	33.0	32.4		28.8	22.5	
Net current transfers (negative = inflow)	-7.6	-4.8	-3.9	-4.0	1.7	-2.9	-2.3	-2.1	-1.9	-1.8	-1.6		-1.4	-1.1	-1.3
Other current account flows (negative = net inflow)	0.6	0.8	1.2			0.9	0.9	1.2	13.0	14.3	13.1		8.5	3.8	
Net FDI (negative = inflow)	-5.2	-6.5	-7.7	-5.2	1.6	-15.4	-18.2	-15.9	-5.7	-5.7	-5.7		-5.5	-5.3	-5.3
Endogenous debt dynamics [2]	22.4	-10.0	-5.6			-1.2	-1.4	-1.5	-2.2	-2.0	-1.5		-1.6	-1.4	
Contribution from nominal interest rate	1.2	0.8	0.3			0.3	0.2	0.2	0.2	0.2	0.3		0.3	0.2	
Contribution from real GDP growth	-5.5	-3.2	-3.1			-1.4	-1.6	-1.8	-2.4	-2.3	-1.8		-1.9	-1.6	
Contribution from price and exchange rate changes	26.6	-7.7	-2.8			…	…	…	…	…	…		…	…	
Residual (3-4) [3]	-28.3	-3.8	-36.2			-2.6	-1.7	-0.2	-0.1	-0.5	-0.6		0.1	1.6	
o/w exceptional financing	-1.5	-1.0	-0.4			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt [4]	…	…	11.5			11.4	11.8	12.9	13.6	14.3	14.9		16.2	12.7	
In percent of exports	…	…	38.7			46.3	54.1	61.0	37.7	38.5	42.3		55.5	60.8	
NPV of PPG external debt	…	…	11.5			11.4	11.8	12.9	13.6	14.2	14.9		16.2	12.7	
In percent of exports	…	…	38.7			46.2	54.1	60.9	37.7	38.5	42.2		55.5	60.7	
Debt service-to-exports ratio (percent)	107.0	8.3	3.5			1.6	1.7	1.7	0.9	0.9	1.2		2.2	2.7	
PPG debt service-to-exports ratio (percent)	107.0	8.3	3.5			1.6	1.7	1.7	0.9	0.9	1.2		2.2	2.7	
Total gross financing need (billions of U.S. dollars)	1639.4	294.5	102.9			158.3	245.5	320.8	326.4	374.8	421.2		387.4	-218.9	
Non-interest current account deficit that stabilizes debt ratio	11.2	20.3	49.5			19.2	21.2	17.7	8.0	8.2	7.7		7.0	5.1	
Key macroeconomic assumptions															
Real GDP growth (percent)	5.3	4.6	4.9	3.5	6.7	6.5	7.3	7.5	9.8	9.0	7.2	7.9	7.1	7.2	7.1
GDP deflator in US dollar terms (percent change)	-24.2	10.6	4.2	2.5	12.6	20.5	12.4	2.8	2.7	2.9	2.4	7.3	2.3	2.4	2.3
Effective interest rate (percent) [5]	1.2	1.2	0.4	1.1	0.3	1.1	1.0	1.0	0.9	0.8	1.3	1.0	1.2	0.9	1.1
Growth of exports of G&S (U.S. dollar terms, percent)	12.8	-4.7	20.8	14.0	32.9	6.7	6.3	7.4	92.2	15.0	4.9	22.1	5.6	6.5	5.8
Growth of imports of G&S (U.S. dollar terms, percent)	18.1	-0.1	9.1	12.4	28.3	38.0	20.2	3.6	-8.3	10.7	7.7	12.0	6.8	5.7	7.0
Grant element of new public sector borrowing (percent)	…	…	…	…	…	43.8	44.2	44.2	44.2	44.2	44.2	44.1	44.2	44.2	44.2
Memorandum item:															
Nominal GDP (millions of U.S. dollars)	4357.7	5041.9	5509.8			7067.4	8518.7	9412.9	10615.1	11912.8	13074.5		20676.6	51853.2	

Source: Staff simulations.

[1] Includes both public and private sector external debt.
[2] Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
[3] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
[4] Assumes that NPV of private sector debt is equivalent to its face value.
[5] Current-year interest payments devided by previous period debt stock.
[6] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 1b. Madagascar: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27
(Percent)

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of debt-to-GDP ratio								
Baseline	11	12	13	14	14	15	16	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 [1]	11	12	12	12	11	11	10	13
A2. New public sector loans on less favorable terms in 2007-26 [2]	11	13	15	16	18	19	23	21
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	11	13	16	17	18	18	20	16
B2. Export value growth at historical average minus one standard deviation in 2007-08 [3]	11	15	21	21	21	21	21	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	11	15	18	19	20	21	23	18
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 [4]	11	20	27	27	26	27	24	15
B5. Combination of B1-B4 using one-half standard deviation shocks	11	16	22	23	24	25	26	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 [5]	11	16	18	19	19	20	22	17
NPV of debt-to-exports ratio								
Baseline	46	54	61	38	38	42	55	61
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 [1]	46	54	55	32	31	31	34	61
A2. New public sector loans on less favorable terms in 2007-26 [2]	46	59	70	45	48	54	78	101
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	46	54	61	38	38	42	55	61
B2. Export value growth at historical average minus one standard deviation in 2007-08 [3]	46	89	171	101	99	105	124	117
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	46	54	61	38	38	42	55	61
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 [4]	46	91	128	74	71	75	84	73
B5. Combination of B1-B4 using one-half standard deviation shocks	46	63	88	53	53	58	73	76
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 [5]	46	54	61	38	38	42	55	61
Debt service ratio								
Baseline	2	2	2	1	1	1	2	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 [1]	2	2	3	1	1	2	3	5
A2. New public sector loans on less favorable terms in 2007-26 [2]	2	2	2	2	1	2	3	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	2	2	2	1	1	2	3	4
B2. Export value growth at historical average minus one standard deviation in 2007-08 [3]	2	3	4	3	3	3	7	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	2	2	2	1	1	2	3	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 [4]	2	2	3	2	2	2	5	5
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	3	2	1	2	4	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 [5]	2	2	2	1	1	2	3	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) [6]	42	42	42	42	42	42	42	42

Source: Staff projections and simulations.

[1] Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

[2] Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.

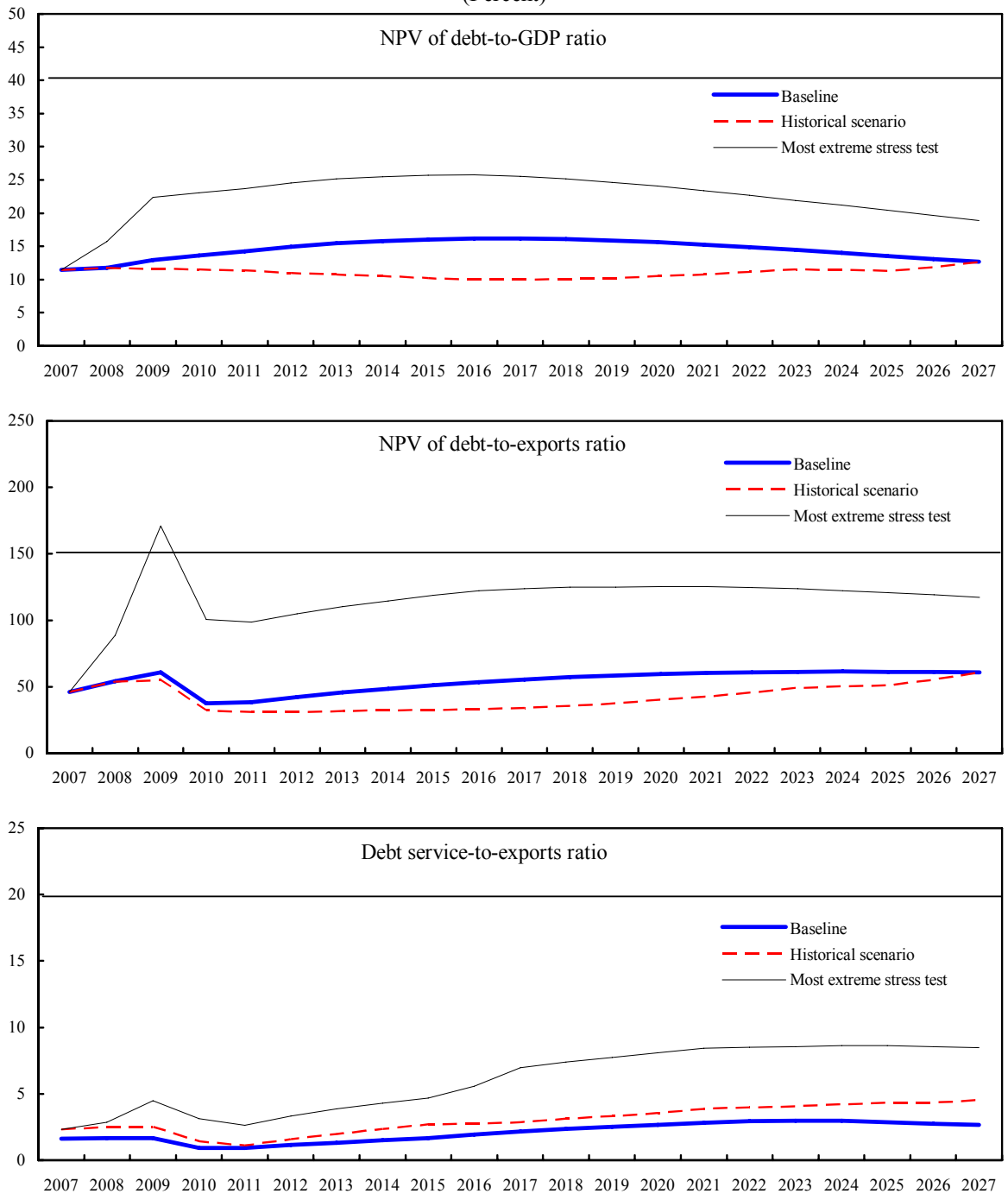
[3] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

[4] Includes official and private transfers and FDI.

[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

[6] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1a. Madagascar: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2007-2027
(Percent)



Source: Staff projections and simulations.

Table 2a.Madagascar: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004-2027
(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average [5]	Standard Deviation [5]	Estimate	Projections									
	2004	2005	2006			2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt [1]	92.4	80.3	38.6			36.4	34.8	35.4	35.0	34.7	35.3		36.6	33.7	
o/w foreign-currency denominated	79.9	69.9	28.8			26.8	26.2	27.5	28.0	28.2	29.0		29.6	23.2	
Change in public sector debt	-6.8	-12.1	-41.7			-2.3	-1.5	0.5	-0.3	-0.4	0.6		0.2	-0.7	
Identified debt-creating flows	10.9	-4.6	-53.7			-2.1	0.8	-0.5	-0.4	-0.5	0.6		0.2	-0.6	
Primary deficit	2.0	2.0	1.1	1.7	1.3	3.8	2.8	2.8	2.8	2.7	2.9	3.0	2.6	1.5	2.3
Revenue and grants	20.3	16.7	18.0			16.1	15.1	15.5	15.9	16.2	16.4		17.2	18.0	
of which: grants	8.2	5.7	6.7			5.0	3.5	3.4	3.3	3.2	3.2		2.9	2.5	
Primary (noninterest) expenditure	22.2	18.7	19.1			19.8	17.9	18.3	18.7	18.9	19.3		19.8	19.6	
Automatic debt dynamics	11.3	-4.9	-12.8			-5.7	-2.0	-3.3	-3.1	-3.1	-2.3		-2.5	-2.2	
Contribution from interest rate/growth differential	-5.8	-5.0	-3.6			-1.8	-2.0	-2.1	-2.9	-2.3	-1.7		-1.8	-1.8	
of which: contribution from average real interest rate	-0.8	-0.9	0.1			0.5	0.4	0.3	0.2	0.6	0.6		0.6	0.5	
of which: contribution from real GDP growth	-5.0	-4.1	-3.7			-2.3	-2.5	-2.4	-3.1	-2.9	-2.3		-2.4	-2.3	
Contribution from real exchange rate depreciation	17.1	0.1	-9.2			-3.9	0.0	-1.2	-0.2	-0.7	-0.6		...	...	
Other identified debt-creating flows	-2.4	-1.7	-42.0			-0.2	-0.1	-0.1	-0.1	-0.1	0.0		0.0	0.0	
Privatization receipts (negative)	-0.4	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-2.0	-1.7	-41.9			-0.1	-0.1	-0.1	-0.1	-0.1	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-17.7	-7.5	12.0			-0.2	-2.3	1.1	0.1	0.1	0.0		0.0	0.0	
NPV of public sector debt	...	...	21.4			21.0	20.4	20.7	20.7	20.7	21.3		23.1	23.1	
o/w foreign-currency denominated	...	...	11.5			11.4	11.8	12.9	13.6	14.2	14.9		16.2	12.7	
o/w external	...	...	11.5			11.4	11.8	12.9	13.6	14.2	14.9		16.2	12.7	
NPV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need [2]	20.5	16.5	13.4			14.2	12.8	11.7	10.7	9.9	9.6		9.6	12.3	
NPV of public sector debt-to-revenue ratio (in percent) [3]	...	...	118.9			130.7	134.9	133.7	129.8	127.4	129.3		134.4	128.4	
o/w external	...	...	64.2			71.2	78.1	83.3	85.5	87.8	90.9		94.1	70.2	
Debt service-to-revenue ratio (in percent) [3, 4]	25.9	26.4	18.9			12.3	10.6	8.8	7.3	6.4	6.1		7.0	8.0	
Primary deficit that stabilizes the debt-to-GDP ratio	8.7	14.1	42.8			6.1	4.4	2.3	3.1	3.0	2.3		2.5	2.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.3	4.6	4.9	3.5	5.9	6.5	7.3	7.5	9.8	9.0	7.2	7.9	7.1	7.2	7.1
Average nominal interest rate on forex debt (in percent)	2.0	1.5	0.8	1.7	0.8	1.0	1.0	1.0	0.9	0.8	1.4	1.0	1.2	0.9	1.1
Average real interest rate on domestic currency debt (in percent)	-2.3	-1.5	9.5	0.7	5.6	8.5	7.8	7.3	6.5	5.5	4.7	6.7	4.6	4.5	4.5
Real exchange rate depreciation (in percent, + indicates depreciation)	21.7	0.1	-14.0	1.4	11.4	-14.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	14.3	18.4	11.3	10.2	4.7	9.5	6.4	5.6	4.9	5.0	4.9	6.1	4.8	4.9	4.9
Growth of real primary spending (deflated by GDP deflator, in percent)	34.7	-12.1	7.0	9.2	21.2	10.7	-3.0	9.8	12.1	10.2	9.6	8.2	7.5	6.6	7.2
Grant element of new external borrowing (in percent)	0.0	43.2	44.2	8.7	18.4	44.2	44.2	44.2	44.2	44.2	44.2	44.2	44.2	45.2	...

Sources: Malagasy authorities; and Fund staff estimates and projections.

[1] Central government. Gross public debt.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues including grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Madagascar: Sensitivity Analysis for Key Indicators of Public Debt 2007-2027

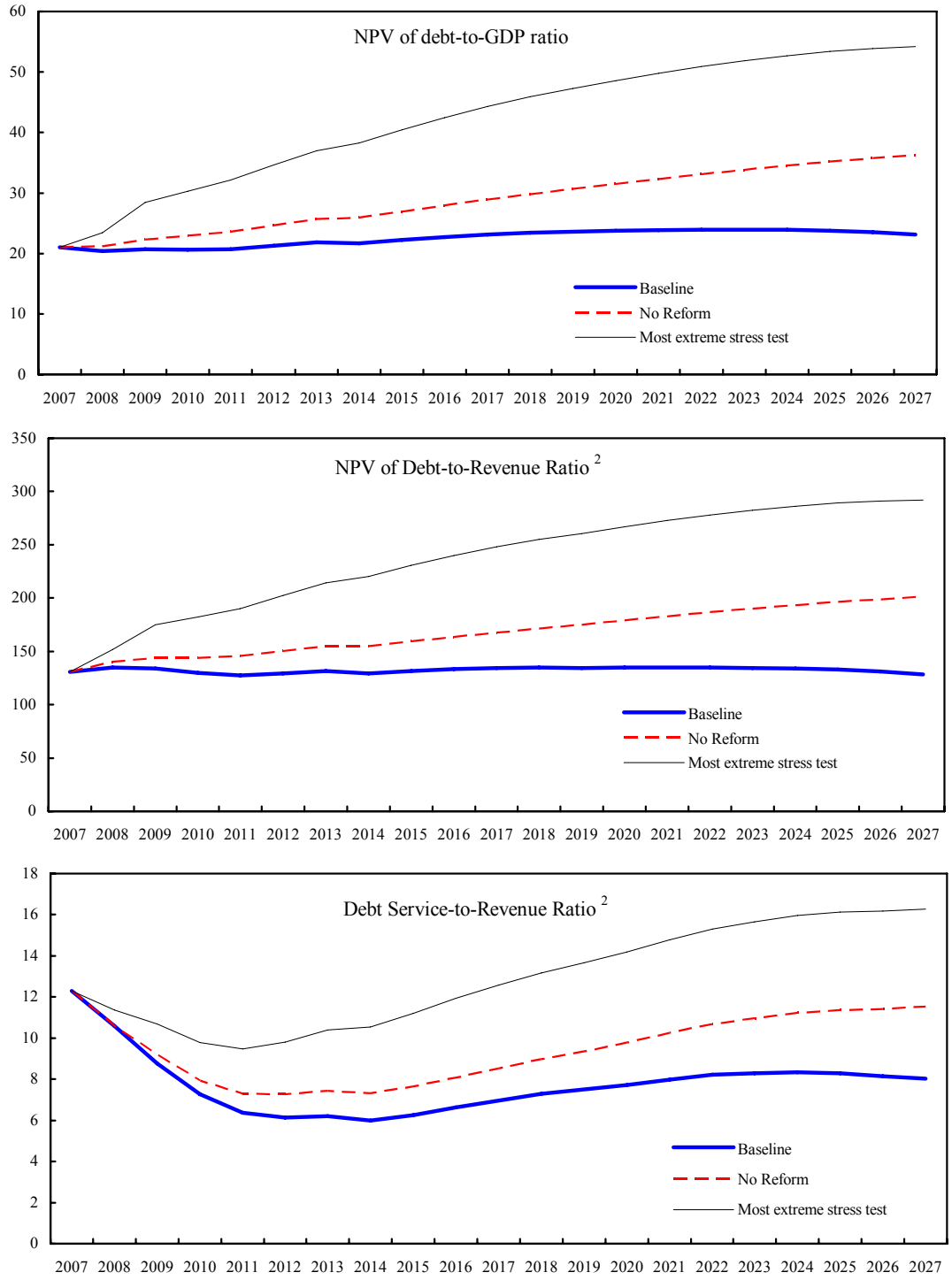
	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Baseline	21	20	21	21	21	21	23	23
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	21	20	20	20	20	20	21	27
A2. Primary balance is unchanged from 2007	21	21	22	23	24	25	29	36
A3. Permanently lower GDP growth [1]	21	21	22	22	23	25	33	54
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	21	23	28	30	32	35	44	54
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	21	20	21	21	21	21	23	23
B3. Combination of B1-B2 using one half standard deviation shocks	21	21	22	22	21	21	22	20
B4. One-time 30 percent real depreciation in 2008	21	25	24	23	22	23	23	23
B5. 10 percent of GDP increase in other debt-creating flows in 2008	21	29	29	28	27	27	27	25
NPV of Debt-to-Revenue Ratio [2]								
Baseline	131	135	134	130	127	129	134	128
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	131	131	127	123	120	119	115	130
A2. Primary balance is unchanged from 2007	131	140	144	144	146	150	168	201
A3. Permanently lower GDP growth 1/	131	137	139	138	140	147	188	288
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	131	152	175	182	190	202	248	292
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	131	136	135	131	128	130	135	129
B3. Combination of B1-B2 using one half standard deviation shocks	131	138	139	132	126	126	122	111
B4. One-time 30 percent real depreciation in 2008	131	167	155	145	138	137	134	126
B5. 10 percent of GDP increase in other debt-creating flows in 2008	131	195	185	173	165	163	156	138
Debt Service-to-Revenue Ratio [2]								
Baseline	12	11	9	7	6	6	7	8
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	12	11	9	7	6	6	6	10
A2. Primary balance is unchanged from 2007	12	11	9	8	7	7	9	12
A3. Permanently lower GDP growth [1]	12	11	9	8	7	7	10	16
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	12	11	11	10	9	10	13	16
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	12	11	9	7	6	6	7	8
B3. Combination of B1-B2 using one half standard deviation shocks	12	11	9	7	6	6	6	7
B4. One-time 30 percent real depreciation in 2008	12	11	9	8	7	7	8	9
B5. 10 percent of GDP increase in other debt-creating flows in 2008	12	11	13	10	9	8	8	9

Sources: Country authorities; and Fund staff estimates and projections.

[1] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).

[2] Revenues are defined inclusive of grants.

Figure 2a. Madagascar: Indicators of Public Debt Under Alternative Scenarios, 2007-2027 [1]



Source: Staff projections and simulations.

[1] Most extreme stress test is that which yields the highest ratio in 2017.

[2] Revenue including grants.